



Weak observability conditions for exponential and prescribed-time stabilization of non-autonomous bilinear systems

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ABSTRACT

This paper explores the stabilization of abstract non-autonomous bilinear systems, focusing on two distinct forms of stabilization of the closed-loop system; namely the exponential stabilization which consists of converging exponentially fast towards an equilibrium point as time progresses, and prescribed time stabilization, which ensures the convergence within a specific time. We construct a feedback control by computing the time derivative of the system's "energy". Based on the theories of nonlinear maximal monotone operators and fixed point theory, we analyze the well-posedness of the system in closed-loop. Then, using a specific weak observability inequality, we prove the uniform exponential stability of the system in closed loop. Subsequently, We utilize the acquired findings to establish prescribed time stability. Finally, applications to some partial differential equations are given.

1. Introduction

Feedback stabilization represents fundamental challenges in the analysis and design of control systems. The problem of stabilization of autonomous bilinear systems has received considerable attention from numerous researchers employing diverse control methodologies such as quadratic control laws, sliding mode control, piecewise constant feedback and optimal control laws (see, e.g. Ball & Slemrod, 1979; Ouzahra, 2020 and references therein). More recently, the question of stabilizing bilinear systems with unbounded control operators has been addressed in Ammari, El Alaoui, and Ouzahra (2021), Berrahmoune (2010), El Alaoui and Ouzahra (2024), El Ayadi, Ouzahra, and Boutoulout (2012).

However, many real-world applications require time-varying dynamic models, leading to research into nonautonomous systems. These models are particularly prevalent in fields such as automotive industry and aerospace (see Li, Xie, et al., 2019; Xu et al., 2021; Zhou, 2021).

Stabilizing non-autonomous systems poses a fundamental challenge in control theory due to their inherent time-varying dynamics, unlike autonomous systems with static dynamics. This complexity stems from the need to address both uniformity concerns in stability-related issues and the deficiency of mathematical tools comparable to those for autonomous systems. Consequently, the stability analysis and design of time-varying systems require the imposition of additional conditions, prompting the development of specialized methodologies and theoretical frameworks tailored to their dynamic nature.

In recent decades, significant attention has been devoted to the study of finite-dimensional non-autonomous systems, as demonstrated

by works such as Anderson, Ilchmann, and Wirth (2013), Zhang, Han, Tang, and Liu (2019), Zhou (2016) and many related studies. Within this framework, stabilization methodologies have evolved to address the unique challenges posed by time-varying dynamics and uncertainties. Robust control techniques, including H_∞ synthesis and sliding mode control, have been widely adopted to ensure stability under disturbances and modeling inaccuracies. In addition to these, adaptive control strategies dynamically adjust parameters in real time to accommodate unknown or varying system behaviors (Astolfi, Karagiannis, & Ortega, 2008; Chen & Huang, 2005; Xu et al., 2021). Central to the analysis of such systems, Lyapunov-based approaches (Li, Yang, & Song, 2019; Zhang, Zhou, & Duan, 2024) have emerged as indispensable tools, providing rigorous stability guarantees by using energy-like functions tailored to time-varying structures.

In contrast, the stability of non-autonomous systems in infinite-dimensional spaces remains relatively under-explored, with only a few studies addressing cases involving bounded dynamics (Barreira, 2008; Barreira & Valls, 2005) and related contributions.

Regarding the prescribed-time stabilization, which enables faster convergence rate than asymptotic stability by allowing system trajectories to reach zero within a finite time frame, extensive attention has been directed towards autonomous finite-dimensional systems. This area has been thoroughly explored in various studies, including Bhat and Bernstein (2000), Lopez-Ramirez, Efimov, Polyakov, and Perruquetti (2019), Polyakov (2012), Polyakov, Efimov, and Perruquetti

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(2015), Song, Wang, Holloway, and Krstic (2017), Song, Wang, and Krstic (2019).

Similar to the focus on finite-dimensional systems, recent studies have increasingly directed attention towards evolution systems in infinite-dimensional spaces (Jammazi, Boutayeb, & Saidi, 2023; Polyakov, Coron, & Rosier, 2017, 2018) using the concept of homogeneity combined with Lyapunov method in the context of linear systems. The case of bilinear heat equation was studied in Espitia, Polyakov, Efimov, and Perruquetti (2019), Ouzahra (2021). In Sogoré and Jammazi (2020), the finite-time stabilization of abstract bilinear control systems with a bounded control operator is investigated, while Najib and Ouzahra (2024b) explores the unbounded case. Additionally, the output finite-time stability has been recently examined in Najib and Ouzahra (2024a). Recently, a study introduced a result on small-time/prescribed time stability of abstract bilinear systems in infinite dimensions (Jammazi, Ouzahra, & Sogoré, 2023). Additionally, research on finite-time stabilization of non-autonomous systems in finite dimensions has gained attention in recent years (Li, Jiang, Ren, & Zhang, 2022; Moulay & Perruquetti, 2008; Zhou & Shi, 2021).

An analysis of the literature mentioned above reveals a predominant focus on either autonomous systems or non-autonomous systems in finite dimensions. However, in the process of designing mathematical or computational models to represent specific systems or phenomena, the inclusion of assumptions, equations, and constraints may result in the emergence of time-varying dynamics in infinite-dimensional spaces.

Therefore, the primary contribution of this research lies in the study of the stabilization of abstract infinite-dimensional non-autonomous bilinear systems, a topic that remains underexplored in existing literature. While previous works have extensively studied the stabilization of autonomous bilinear systems and finite-dimensional non-autonomous systems, our study addresses a significant gap by investigating two distinct forms of stability for non-autonomous bilinear systems in infinite dimensions:

1. **Uniform Exponential Stabilization:** We develop a feedback control strategy based on a Lyapunov function and a weak observability inequality to ensure uniform exponential stabilization. The well-posedness of the closed-loop system is demonstrated by applying the theory of nonlinear maximal monotone operators and fixed point arguments.
2. **Prescribed-Time Stabilization:** We establish a control law that ensures stabilization within a prescribed, finite-time horizon, independent of the initial condition, by performing a variable change to reduce it to the previous case, i.e., the asymptotic case.

Additionally, we apply our theoretical findings to practical examples, including the stabilization of non-autonomous diffusion-reaction and transport equations.

The paper is structured as follows: In Section 2, we provide results about uniform exponential stability of non-autonomous systems under bilinear controls. Section 3 delves into the stabilization in prescribed time of bilinear non-autonomous systems. Section 4 focuses on some examples of non-autonomous diffusion-reaction equation and transport equation.

2. Uniform exponential stabilization

Let us consider the following non-autonomous bilinear system:

$$\begin{cases} \dot{y} = A(t)y + u(t, y)B(t)y, & t \geq t_0 \geq 0 \\ y(t_0) = y_0 \in H, \end{cases} \quad (2.1)$$

where $\{A(t)\}_{t \geq t_0}$ is a family of linear dissipative operators. H denotes a real Hilbert space equipped with an inner product $\langle \cdot, \cdot \rangle$ and norm $\|\cdot\|$. The system state is represented by $y(t)$. The function $u : \mathbb{R}^+ \times H \rightarrow \mathbb{R}$ corresponds to a scalar (time-varying) feedback control. The family of linear operator $\{B(t)\}_{t \geq t_0}$ is bounded from H to H and there exists

$L \in L^\infty(t_0, +\infty)$ such that for all $y \in H$, we have $\|B(t)y\| \leq L(t)\|y\|$, $\forall t \geq t_0$ (i.e. $\|B(\cdot)\| \in L^\infty(t_0, +\infty)$).

We assume the existence of the Gelfand triple $V \subset H \equiv H' \subset V'$, where V is a Hilbert space and V' its dual w.r.t the pivot space H , ensuring that $\langle v, w \rangle_{V', V} = \langle v, w \rangle$ for all $v, w \in H$. Furthermore, we assume that the operators $\{A(t)\}_{t \geq t_0}$ are bounded from V to V' in the sense that for any $T > t_0$ there exists a constant $M = M(T, t_0)$ such that $|\langle A(t)y, z \rangle_{V', V}| \leq M\|y\|_V\|z\|_V$, for all $y, z \in V$;

In the sequel, we further assume that, for $T > t_0$,

- (H_1). for every $y, z \in V$, the function $t \rightarrow \langle A(t)y, z \rangle_{V', V}$ is measurable.
- (H_2). There exist $\alpha_T = \alpha_T(T) > 0$ and $\lambda_T = \lambda_T(T) \in \mathbb{R}$ such that $\langle -A(t)y, y \rangle_{V', V} + \lambda_T\|y\|_V^2 \geq \alpha_T\|y\|_V^2$, $\forall y \in V$, $\forall t \in [t_0, T]$.

Remark 2.1.

- The assumption (H_1) guarantees the structured evolution of the system over time, while the assumption (H_2) prevents energy blow-up, ensuring that solutions exist globally in time.
- Let us note that assumptions (H_1) and (H_2) will be only used to prove the well-posedness of the system (2.1). Therefore, once the well-posedness is guaranteed, these assumptions become dispensable.

2.1. A well-posedness result

Let us prove the following existence and uniqueness result regarding system (2.1). The proof is based on a fixed point argument.

Theorem 2.1. *Let $T > t_0$ and assume that (H_1) and (H_2) hold, and that the function defined by $F(t, y) = u(t, y)B(t)y$ satisfies*

$$\begin{aligned} \exists K > 0 / \|F(t, y_1) - F(t, y_2)\| &\leq K\|y_1 - y_2\|, \\ \text{for all } y_1, y_2 \in H \text{ and } t \in [t_0, T]. \end{aligned} \quad (2.2)$$

Then for every $y_0 \in H$, the system (2.1) admits a unique weak solution y such that

$$y \in W(t_0, T; V, V') = \{u/u \in L^2(t_0, T; V), u' \in L^2(t_0, T; V')\}.$$

Proof. Step 1: Wellposedness of the auxiliary system: For any $\xi \in C := C([t_0, T]; H)$; the space of continuous function from $[t_0, T]$ to H , we consider the following auxiliary system:

$$\begin{cases} \dot{y}_\xi = A(t)y_\xi + u(t, \xi)B(t)\xi, & t \in [t_0, T] \\ y(t_0) = y_0 \in H \end{cases} \quad (2.3)$$

Taking into account the above assumptions, we deduce from (Dautray & Lions, 1992, p. 512, Theorem 1 & p. 513, Theorem 2), that the system (2.3) admits a unique solution $y_\xi \in W(t_0, T; V, V')$ for any $\xi \in C$.

Step 2: Fixed point argument: Taking into account that $W(t_0, T; V, V') \subset C$ (see Dautray & Lions, 1992, p. 473, Theorem 1), we consider the following map

$$\Phi : C \rightarrow C$$

$$\xi \mapsto y_\xi$$

and we show that Φ is a contraction mapping.

Let $\xi_1, \xi_2 \in C$. We have

$$\|\Phi(\xi_1) - \Phi(\xi_2)\|_C = \sup_{t \in [t_0, T]} \|y_{\xi_1}(t) - y_{\xi_2}(t)\| \quad (2.4)$$

Let $\Psi = y_{\xi_1} - y_{\xi_2}$, which is the solution of the following system:

$$\begin{cases} \dot{\Psi} = A(t)\Psi + u(t, \xi_1)B(t)\xi_1 - u(t, \xi_2)B(t)\xi_2, & t \in [t_0, T] \\ \Psi(t_0) = 0 \in H \end{cases} \quad (2.5)$$

Multiplying the system (2.5) by Ψ , and taking into account the dissipativity of $A(t)$ we deduce that

$$\begin{aligned} \frac{1}{2} \frac{d}{dt} \|\Psi(t)\|^2 &\leq \langle u(t, \xi_1(t))B(t)\xi_1(t) - u(t, \xi_2(t))B(t)\xi_2(t), \Psi(t) \rangle \\ &\leq \|u(t, \xi_1(t))B(t)\xi_1(t) - u(t, \xi_2(t))B(t)\xi_2(t)\| \\ &\quad \times \|\Psi(t)\| \end{aligned}$$

Taking into account that the function $F(t, y) = u(t, y)B(t)y$ satisfies (2.2) we deduce that

$$\frac{1}{2} \frac{d}{dt} \|\Psi(t)\|^2 \leq K \|\xi_1(t) - \xi_2(t)\| \|\Psi(t)\|. \quad (2.6)$$

Integrating (2.6) over $[t_0, t]$, we obtain

$$\begin{aligned} \|\Psi(t)\|^2 &\leq 2K \int_{t_0}^t \|\xi_1(s) - \xi_2(s)\| \|\Psi(s)\| ds \\ &\leq 2K \left(\int_{t_0}^t \|\xi_1(s) - \xi_2(s)\|^2 ds \right)^{\frac{1}{2}} \left(\int_{t_0}^t \|\Psi(s)\|^2 ds \right)^{\frac{1}{2}} \\ &\leq K \left(\int_{t_0}^T \|\xi_1(s) - \xi_2(s)\|^2 ds + \int_{t_0}^t \|\Psi(s)\|^2 ds \right). \end{aligned}$$

Gronwall's lemma then yields

$$\begin{aligned} \|\Psi(t)\|^2 &\leq K \left(\exp(K(T - t_0)) \int_{t_0}^T \|\xi_1(s) - \xi_2(s)\|^2 ds \right) \\ &\leq K(T - t_0) \exp(K(T - t_0)) \|\xi_1 - \xi_2\|_C^2. \end{aligned}$$

That is

$$\begin{aligned} \|\Psi\|_C &= \|\Phi(\xi_1) - \Phi(\xi_2)\|_C \\ &\leq \sqrt{K(T - t_0) \exp(K(T - t_0))} \|\xi_1 - \xi_2\|_C. \end{aligned}$$

Step 3: Wellposedness of the bilinear system:

Let us divide the interval $[t_0, T]$ into N subintervals $[t_0, T_1], \dots, [T_{N-1}, T_N]$ (with $T_N = T$) such that

$$\sqrt{K(T_{j+1} - T_j) \exp(K(T_{j+1} - T_j))} \leq 1/2, \forall j = 0, \dots, N-1.$$

Using the contraction argument above, we prove that there exists a solution of (2.1) on $[t_0, T_1]$. Then, taking $y(T_1)$ as initial condition for problem (2.1), we can iterate the procedure on the second interval $[T_1, T_2]$, and so on up to the final interval $[T_{N-1}, T_N]$.

In other words, considering the following systems for $j = 0, \dots, N-1$

$$(E_j) \begin{cases} \dot{y}_j = A(t)y_j + u(t, y_j)B(t)y_j, & t \in [T_j, T_{j+1}] \\ y(T_j) = y(T_j^-) \in H. \end{cases} \quad (2.7)$$

the corresponding function Φ is a contraction. Hence, we deduce the existence and uniqueness of a solution y_j for (E_j) , $j = 0, \dots, N-1$. So that, the weak solution of (2.1) can then be defined as follows

$$y(t) = y_j(t), \quad \forall t \in [T_j, T_{j+1}].$$

That achieves the proof.

2.2. Sufficient conditions for uniform exponential stability

Let us adopt the following definitions:

Definition 2.1. We say that the system (2.1) is **Lyapunov stabilizable** if there exists a **state feedback control** $u = u(t, y)$ such that for each $t_0 \geq 0$ and for every $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon, t_0) > 0$ such that for any initial condition $y_0 \in H$, if $\|y_0\| \leq \delta$, then the corresponding controlled solution $y(t)$ satisfies:

$$\|y(t)\| \leq \varepsilon, \quad \forall t \geq t_0.$$

If in addition the feedback control u ensures that there exist constants $M, \sigma > 0$ (which may depend on t_0 and y_0) such that the solution satisfies the **exponential decay estimate**:

$$\|y(t)\| \leq M e^{-\sigma(t-t_0)} \|y_0\|, \quad \forall t \geq t_0 \geq 0, \quad (2.8)$$

then the system is called **exponentially stabilizable**.

Furthermore, if the estimate (2.8) holds with constants $M, \sigma > 0$ that are **independent** of t_0 and y_0 , then the system is called **uniformly exponentially stabilizable**.

Remark 2.2.

- In the above definitions, the existence of the feedback control $u = u(t, y)$ ensures that the system is not only stable but also exhibits attractiveness, meaning that trajectories of the system are driven toward the desired equilibrium state over time.
- It is worth noting that if the system is dissipative, it inherently possesses Lyapunov stability, as dissipativity ensures boundedness of system trajectories over time.

2.2.1. A weak observability condition

In this part, we suppose that the family of linear operators $\{B(t)_{t \geq t_0}\}$ is continuous and satisfies the following observability inequality:

$$\exists \tau, \delta > 0, \quad \int_{t_1}^{t_1+\tau} |\langle B(t)\varphi_{t_1}(t), \varphi_{t_1}(t) \rangle| dt \geq \delta \|\varphi_{t_1}(t_1 + \tau)\|^2, \quad (B)$$

$$\forall t_1 \geq t_0, \quad \forall \varphi_1 \in H$$

where φ_{t_1} is the weak solution of the homogeneous system

$$\begin{cases} \dot{\varphi}_{t_1}(t) = A(t)\varphi_{t_1}(t), & t \geq t_1 \\ \varphi_{t_1}(t_1) = \varphi_1. \end{cases} \quad (2.9)$$

Remark 2.3. The weak observability inequality plays a fundamental role in the stabilization of non-autonomous bilinear systems. It ensures that the state remains sufficiently observable over a finite time interval, which is a crucial property for the design of effective feedback controls. In particular, this inequality establishes a direct link between the system's inherent dynamics and the ability of the control law $u(t, y) = -\frac{\langle B(t)y, y \rangle}{\|y\|^2}$ to drive the state to zero exponentially.

In this theorem, we explore uniform exponential stabilization using energy-based methods, which is inspired by the autonomous case.

Theorem 2.2. Assume that assumptions (H_1) , (H_2) and (B) hold. Then, for all $y_0 \in H$, system (2.1) under the following feedback:

$$u(t, y) = \begin{cases} -\frac{\langle B(t)y, y \rangle}{\|y\|^2}, & \text{if } y \neq 0 \\ 0, & \text{otherwise} \end{cases} \quad (2.10)$$

is **uniformly exponentially stable**.

Proof.

Step 1: Wellposedness of the closed-loop system: Observing that $F(t, y) = -\frac{\langle B(t)y, y \rangle}{\|y\|^2} B(t)y$ satisfies (2.2) (with $K = 5\|B\|$), we deduce from Theorem 2.1 that the system (2.1) under the control (2.10) admits a unique weak solution.

Step 2: The estimate $\|y - \varphi_{t_1}\|$ Now, let us consider $z(t) = y(t) - \varphi_{t_1}(t)$ with $\varphi_{t_1}(t_1) = y(t_1)$, then z is the solution of the following system:

$$\begin{cases} \dot{z} = A(t)z + u(t, y)B(t)y, & t \geq t_1 \\ z(t_1) = 0 \end{cases} \quad (2.11)$$

We have

$$\begin{aligned} \langle B(t)\varphi_{t_1}(t), \varphi_{t_1}(t) \rangle &= \langle B(t)(\varphi_{t_1}(t) - y(t)), \varphi_{t_1}(t) \rangle \\ &\quad + \langle B(t)y(t), \varphi_{t_1}(t) - y(t) \rangle \\ &\quad + \langle B(t)y(t), y(t) \rangle, \end{aligned}$$

from which, we can deduce that

$$|\langle B(t)\varphi_{t_1}(t), \varphi_{t_1}(t) \rangle| \leq 2\|B\| \|z(t)\| \|\varphi_{t_1}(t)\| + |\langle B(t)y(t), y(t) \rangle|,$$

where $\|B\| = \sup_{t \geq t_0} \|B(t)\|$.

Integrating over $[t_1, t_1 + \tau]$, we obtain, using the monotonic decrease of the function $t \mapsto \|y(t)\|$ (since for $V = \frac{1}{2}\|y(t)\|^2$, the time derivative satisfies $\dot{V} \leq 0$, implying that V is decreasing and thus $\|y(t)\|$ as well).

$$\int_{t_1}^{t_1+\tau} |\langle B(t)\varphi_{t_1}(t), \varphi_{t_1}(t) \rangle| dt \leq 2\|B\|\|y(t_1)\| \int_{t_1}^{t_1+\tau} \|z(t)\| dt + \int_{t_1}^{t_1+\tau} |\langle B(t)y(t), y(t) \rangle| dt. \quad (2.12)$$

On one hand, multiplying the system (2.11) by z and considering the dissipativity of $A(t)$, we derive that

$$\frac{1}{2} \frac{d}{dt} \|z(t)\|^2 \leq |u(t, y(t))| |\langle B(t)y(t), z(t) \rangle| \leq |u(t, y(t))| \|B\| \|y(t)\| \|z(t)\|. \quad (2.13)$$

Integrating over $[t_1, t]$ with $t_1 \leq t \leq t_1 + \tau$, we deduce that

$$\begin{aligned} \|z(t)\|^2 &\leq 2 \int_{t_1}^t \|B\| |u(s, y(s))| \|y(s)\| \|z(s)\| ds \\ &\leq 2 \left(\int_{t_1}^t \|B\|^2 |u(s, y(s))|^2 \|y(s)\|^2 ds \right)^{\frac{1}{2}} \\ &\times \left(\int_{t_1}^t \|z(s)\|^2 ds \right)^{\frac{1}{2}} \\ &\leq \int_{t_1}^t \|B\|^2 |u(s, y(s))|^2 \|y(s)\|^2 ds + \int_{t_1}^t \|z(s)\|^2 ds \\ &\leq \int_{t_1}^{t_1+\tau} \|B\|^2 |u(s, y(s))|^2 \|y(s)\|^2 ds + \int_{t_1}^t \|z(s)\|^2 ds \end{aligned} \quad (2.14)$$

Gronwall's lemma then yields

$$\|z(t)\|^2 \leq \exp(\tau) \|B\|^2 \left(\int_{t_1}^{t_1+\tau} |u(s, y(s))|^2 \|y(s)\|^2 ds \right) \quad (2.15)$$

Taking into account the decreasing of y we deduce that

$$\|z(t)\| \leq \exp\left(\frac{\tau}{2}\right) \|B\| \|y(t_1)\| \left(\int_{t_1}^{t_1+\tau} |u(s, y(s))|^2 ds \right)^{\frac{1}{2}} \quad (2.16)$$

Step 2: Estimate $\|y(t_1 + \tau)\|$ under observability condition It follows that

$$\begin{aligned} \|y(t_1 + \tau)\|^2 &\leq 2\|\varphi_{t_1}(t_1 + \tau)\|^2 \\ &+ 2\exp(\tau) \|B\|^2 \|y(t_1)\|^2 \int_{t_1}^{t_1+\tau} |u(s, y(s))|^2 ds \\ &\leq 2\|\varphi_{t_1}(t_1 + \tau)\|^2 \\ &+ 2\tau \exp(\tau) \|B\|^3 \|y(t_1)\|^2 \left(\int_{t_1}^{t_1+\tau} |u(s, y(s))|^2 ds \right)^{\frac{1}{2}} \end{aligned} \quad (2.17)$$

On the other hand, we have

$$\begin{aligned} \int_{t_1}^{t_1+\tau} |\langle B(s)y(s), y(s) \rangle| ds &\leq \int_{t_1}^{t_1+\tau} |u(s, y(s))| \|y(s)\|^2 ds \\ &\leq \|y(t_1)\|^2 \int_{t_1}^{t_1+\tau} |u(s, y(s))| ds \\ &\leq \tau \|y(t_1)\|^2 \\ &\times \left(\int_{t_1}^{t_1+\tau} |u(s, y(s))|^2 ds \right)^{\frac{1}{2}}. \end{aligned} \quad (2.18)$$

Now, combining (2.12), (2.15) and (2.18), we deduce that

$$\begin{aligned} \int_{t_1}^{t_1+\tau} |\langle B(t)\varphi_{t_1}(t), \varphi_{t_1}(t) \rangle| dt &\leq 2\tau e^{\frac{\tau}{2}} \|B\|^2 \|y(t_1)\|^2 \\ &\times \left(\int_{t_1}^{t_1+\tau} |u(s, y(s))|^2 ds \right)^{\frac{1}{2}} \\ &+ \tau \|y(t_1)\|^2 \left(\int_{t_1}^{t_1+\tau} |u(s, y(s))|^2 ds \right)^{\frac{1}{2}} \\ &\leq (2\tau e^{\frac{\tau}{2}} \|B\|^2 + \tau) \|y(t_1)\|^2 \\ &\times \left(\int_{t_1}^{t_1+\tau} |u(s, y(s))|^2 ds \right)^{\frac{1}{2}}. \end{aligned} \quad (2.19)$$

This together with (2.17) and (B) implies

$$\begin{aligned} \frac{\delta}{2} \|y(t_1 + \tau)\|^2 &\leq (2\tau \exp(\frac{\tau}{2}) \|B\|^2 + \tau) \|y(t_1)\|^2 \\ &\times \left(\int_{t_1}^{t_1+\tau} |u(s, y(s))|^2 ds \right)^{\frac{1}{2}} \\ &+ \delta \tau \exp(\tau) \|B\|^3 \|y(t_1)\|^2 \\ &\times \left(\int_{t_1}^{t_1+\tau} |u(s, y(s))|^2 ds \right)^{\frac{1}{2}}. \end{aligned}$$

This yields

$$\|y(t_1 + \tau)\|^2 \leq \sqrt{2} a \|y(t_1)\|^2 \left(\int_{t_1}^{t_1+\tau} |u(s, y(s))|^2 ds \right)^{\frac{1}{2}}, \quad (2.20)$$

where $a = \frac{\sqrt{2}}{\delta} \left(\delta \tau \exp(\tau) \|B\|^3 + 2\tau \exp(\frac{\tau}{2}) \|B\|^2 + \tau \right) > 0$.

Multiplying system (2.1)–(2.10) by $y(t)$ and integrating over the interval $[t_1, t_1 + \tau]$, we obtain, via the dissipativity of $A(t)$ that

$$\begin{aligned} \|y(t_1 + \tau)\|^2 - \|y(t_1)\|^2 &\leq 2 \int_{t_1}^{t_1+\tau} u(t, y(t)) \langle B(t)y(t), y(t) \rangle dt \\ &\leq -2 \int_{t_1}^{t_1+\tau} u(t, y(t))^2 \|y(t)\|^2 dt \\ &\leq -2 \|y(t_1 + \tau)\|^2 \int_{t_1}^{t_1+\tau} u(t, y(t))^2 dt \end{aligned} \quad (2.21)$$

Then, using (2.20), we derive from (2.21)

$$\begin{aligned} \|y(t_1)\|^4 \left(\|y(t_1 + \tau)\|^2 - \|y(t_1)\|^2 \right) &\leq -2 \|y(t_1)\|^4 \|y(t_1 + \tau)\|^2 \\ &\times \int_{t_1}^{t_1+\tau} u(t, y(t))^2 dt \\ &\leq -\frac{1}{a^2} \|y(t_1 + \tau)\|^6 \end{aligned}$$

Step 3: Exponential decay Now, let us consider the following sequence $V_k = \|y(k\tau)\|^2$, $k \in \mathbb{N}$. Then the last inequality implies

$$V_k^2 (V_k - V_{k+1}) \geq \frac{1}{a^2} V_{k+1}^3$$

Since V_k is positively decreasing, we deduce that

$$\begin{aligned} \left(V_k^{\frac{3}{2}} - \frac{1}{2} V_{k+1}^{\frac{3}{2}} \right)^2 &= V_k^3 - V_k^{\frac{3}{2}} V_{k+1}^{\frac{3}{2}} + \frac{1}{4} V_{k+1}^3 \\ &\geq V_k^3 - V_k^2 V_{k+1} + \frac{1}{4} V_{k+1}^3 \\ &\geq \left(\frac{1}{a^2} + \frac{1}{4} \right) V_{k+1}^3. \end{aligned}$$

Then we conclude that

$$V_k^{\frac{3}{2}} \geq \left(\left(\frac{1}{a^2} + \frac{1}{4} \right)^{\frac{1}{2}} + \frac{1}{2} \right) V_{k+1}^{\frac{3}{2}}, \quad \forall k \in \mathbb{N}$$

and hence, we deduce that $V_{k+1} \leq c V_k$ with $c = \left(\left(\frac{1}{a^2} + \frac{1}{4} \right)^{\frac{1}{2}} + \frac{1}{2} \right)^{-\frac{2}{3}} < 1$, from which, we derive that $V_k \leq c^{k-k_0} \|y(t_0)\|^2$ with $k_0 = E(t_0/\tau) + 1$ (where $E(\cdot)$ is the integer part). We conclude, by using this last estimate of V_k and the monotony of $\|y(t)\|$, that for some $M, \sigma > 0$ we have

$$\|y(t)\|^2 \leq M e^{-\sigma(t-t_0)} \|y_0\|^2 \quad (2.22)$$

2.2.2. Necessary conditions for uniform exponential stability

In the next theorem, we give a necessary condition for the uniform exponential stabilization of (2.1) under the control (2.10). Let us introduce the following assumption:

(H₃). For all $t \geq 0$, we have $\langle A(t)y, y \rangle_{V', V} = 0$, $\forall y \in V$.

Theorem 2.3.

Let assumption (H₃) hold. If the system (2.1) is uniformly exponentially stabilizable under the control (2.10), then the estimate (B) holds.

Proof. Let $t_1 \geq 0$, $\varphi \in H$ and let us consider the system (2.1) with $y_0 = \varphi$ at $t_0 = t_1$.

The fact that system (2.1) is uniformly exponentially stabilizable, gives that

$$\exists M > 1, \omega > 0 \quad \|y(t)\|^2 \leq M e^{-\omega(t-t_1)} \|\varphi\|^2, \quad \forall t \geq t_1.$$

Multiplying system (2.1) by y and integrating over the interval $[t_1, t_1 + \tau]$ with $\tau < \frac{\ln(M)}{\omega}$, we obtain, via (H₃)

$$\begin{aligned} \int_{t_1}^{t_1+\tau} \langle -u(s, y(s))B(s)y(s), y(s) \rangle ds &= \frac{1}{2} (\|\varphi\|^2 - \|y(t_1 + \tau)\|^2) \\ &\geq \frac{1}{2} (1 - M e^{-\omega\tau}) \|\varphi\|^2. \end{aligned}$$

Moreover, by Cauchy–Schwarz inequality, we obtain

$$\begin{aligned} \int_{t_1}^{t_1+\tau} \langle -u(s, y(s))B(s)y(s), y(s) \rangle ds \\ \leq \int_{t_1}^{t_1+\tau} \|u(s, y(s))B(s)y(s)\| \|y(s)\| ds \\ \leq \tau \|\varphi\| \left(\int_{t_1}^{t_1+\tau} \|u(s, y(s))B(s)y(s)\|^2 ds \right)^{\frac{1}{2}}. \end{aligned}$$

We deduce from the two last inequalities that

$$\begin{aligned} \int_{t_1}^{t_1+\tau} \|u(s, y(s))B(s)y(s)\|^2 ds &\geq \left(\frac{1 - M e^{-\omega\tau}}{2\tau} \right)^2 \|\varphi\|^2 \\ &\geq \left(\frac{1 - M e^{-\omega\tau}}{2\tau} \right)^2 \|\varphi_{t_1}(t_1 + \tau)\|^2. \end{aligned} \quad (2.23)$$

Now, let us set $z(t) = y(t) - \varphi_{t_1}(t)$, which is a solution of the following system:

$$\begin{cases} \dot{z}(t) = A(t)z(t) + u(t, y(t))B(t)y(t), & t \geq t_1 \\ z(t_1) = 0 \end{cases}$$

Multiplying the last system by $z(s)$, for $t_1 \leq s \leq t \leq t_1 + \tau$, and integrating over the interval $[t_1, s]$ yields

$$\begin{aligned} \|z(s)\|^2 &\leq 2 \int_{t_1}^s \|u(r, y(r))B(r)y(r)\| \|z(r)\| dr \\ &\leq 2 \left(\int_{t_1}^s \|u(r, y(r))B(r)y(r)\|^2 dr \right)^{\frac{1}{2}} \left(\int_{t_1}^s \|z(r)\|^2 dr \right)^{\frac{1}{2}} \\ &\leq \int_{t_1}^s \|u(r, y(r))B(r)y(r)\|^2 dr + \int_{t_1}^s \|z(r)\|^2 dr \\ &\leq \int_{t_1}^t \|u(r, y(r))B(r)y(r)\|^2 dr + \int_{t_1}^s \|z(r)\|^2 dr \end{aligned}$$

Applying Gronwall's lemma, we get

$$\|z(t)\|^2 \leq \exp(\tau) \int_{t_1}^t \|u(r, y(r))B(r)y(r)\|^2 dr.$$

Taking into account that for every $t \geq t_0$, $y \rightarrow u(\cdot, y)B(\cdot)y$ is K -lipschitz with $K = 5\|B\|^2$, we derive

$$\begin{aligned} \|u(t, y(t))B(t)y(t) - u(t, \varphi_{t_1}(t))B(t)\varphi_{t_1}(t)\| \\ \leq 5\|B\|^2 e^{\frac{\tau}{2}} \left(\int_{t_1}^t \|u(r, y(r))B(r)y(r)\|^2 dr \right)^{\frac{1}{2}}. \end{aligned}$$

Then, we deduce that

$$\begin{aligned} \|u(t, y(t))B(t)y(t)\|^2 &\leq 2\|u(t, \varphi_{t_1}(t))B(t)\varphi_{t_1}(t)\|^2 \\ &\quad + 10\|B\|^4 e^{\tau} \int_{t_1}^t \|u(r, y(r))B(r)y(r)\|^2 dr \end{aligned}$$

Thus, by applying Gronwall's lemma (see Corduneanu, 1977, pp. 14–15), we get

$$\begin{aligned} \|u(t, y(t))B(t)y(t)\|^2 &\leq 2\|u(t, \varphi_{t_1}(t))B(t)\varphi_{t_1}(t)\|^2 + 20\|B\|^4 e^{\tau} \\ &\quad \times \int_{t_1}^t \|u(r, \varphi_{t_1}(r))B(r)\varphi_{t_1}(r)\|^2 e^{10\tau\|B\|^4 e^{\tau}} dr. \end{aligned}$$

Integrating the last inequality over $[t_1, t_1 + \tau]$, we obtain

$$\begin{aligned} \int_{t_1}^{t_1+\tau} \|u(t, y(t))B(t)y(t)\|^2 dt \\ \leq 2 \int_{t_1}^{t_1+\tau} \|u(t, \varphi_{t_1}(t))B(t)\varphi_{t_1}(t)\|^2 dt \\ + 20\tau\|B\|^4 e^{\tau+10\tau\|B\|^4 e^{\tau}} \int_{t_1}^{t_1+\tau} \|u(t, \varphi_{t_1}(t))B(t)\varphi_{t_1}(t)\|^2 dt \\ \leq \left(2 + 20\tau\|B\|^4 e^{\tau+10\tau\|B\|^4 e^{\tau}} \right) \\ \times \int_{t_1}^{t_1+\tau} \|u(t, \varphi_{t_1}(t))B(t)\varphi_{t_1}(t)\|^2 dt. \end{aligned}$$

From the last inequality, we deduce via (2.23) that

$$\begin{aligned} \int_{t_1}^{t_1+\tau} |\langle B(t)\varphi_{t_1}(t), \varphi_{t_1}(t) \rangle| dt &\geq \frac{\int_{t_1}^{t_1+\tau} \|u(t, \varphi_{t_1}(t))B(t)\varphi_{t_1}(t)\|^2 dt}{\|B\|^3} \\ &\geq \delta \|\varphi_{t_1}(t_1 + \tau)\|^2 \end{aligned}$$

where $\tau > 0$ is such that $\tau < \frac{\ln(M)}{\omega}$ and

$$\delta = \frac{(1 - M e^{-\omega\tau})^2}{4\|B\|^3 \tau^2 (2 + 20\tau\|B\|^4 \exp(\tau + 10\tau\|B\|^4 e^{\tau}))}.$$

Remark 2.4.

- The assumption (H₃) ensures that $A(t)$ does not introduce any intrinsic energy dissipation or amplification, making the stabilization purely dependent on the control mechanism.
- It is worth noting that under the assumption (H₃), the observability condition (B) stands out as optimal since it serves as both a necessary and sufficient condition for achieving uniform exponential stability in the case where $A(t)$ are of isometries for all $t \geq t_0$. In other words, the observability condition (B) is not just a technical assumption but a fundamental requirement for stabilization in systems governed by conservative dynamics.
- It worth noting that the observability condition (B) can be applied to ensure uniform exponential stability of finite-dimensional systems in the context time-varying dynamics. For further discussion on this subject, the reader is referred to Kalman (1960), Silverman and Meadows (1967), Zhang et al. (2019), Zhou (2021) and the references therein.

2.2.3. A sequential condition

Now let us consider the following asymptotic assumption:

For all $t_n \rightarrow +\infty$ and $y_n \rightharpoonup y$ in H such that

$$\langle B(t_n)y_n, y_n \rangle \rightarrow 0 \quad \text{as } n \rightarrow +\infty, \quad \text{we have } y = 0 \quad (B')$$

Theorem 2.4. Assume that assumptions (H₁), (H₂) and (B') hold. Then, for all $y_0 \in H$, the system (2.1) under the control (2.10) is weakly stable.

Proof. Multiplying system (2.1) by y and integrating over the interval $[t_0, t]$ for $t > t_0$, we obtain, via the dissipativity of $A(t)$ that

$$\begin{aligned} \int_{t_0}^t \langle B(s)y(s), y(s) \rangle^2 ds &\leq -\|y_0\|^2 \int_{t_0}^t u(s, y(s)) \langle B(s)y(s), y(s) \rangle ds \\ &\leq \|y_0\|^2 (\|y_0\|^2 - \|y(t)\|^2) \\ &\leq \|y_0\|^4. \end{aligned}$$

Let (t_n) be a sequence such that $t_n \rightarrow +\infty$. Then, it comes from the convergence of the integral $\int_{t_0}^{+\infty} \langle B(s)y(s), y(s) \rangle^2 ds$, that

$$\int_{t_n+t_0}^{t_n+t} \langle B(s)y(s), y(s) \rangle^2 ds \rightarrow 0 \quad \text{as } n \rightarrow +\infty$$

So, we deduce that

$$\lim_{n \rightarrow +\infty} \int_{t_0}^t \langle B(s+t_n)y(s+t_n), y(s+t_n) \rangle^2 ds = 0$$

Hence, there exists a subsequence still denoted by (t_n) , such that for a.e. $t_1 > t_0$

$$\lim_{n \rightarrow +\infty} \langle B(t_1 + t_n)y(t_1 + t_n), y(t_1 + t_n) \rangle = 0$$

Moreover, it follows from the fact that $\|y(t)\| \leq \|y(t_0)\|$, $t \geq t_0$ and the reflexivity of H , that there is a subsequence of t_n (still denoted t_n) and $y \in H$ such that

$$y(t_1 + t_n) \rightharpoonup y \in H \quad \text{as } t_n \rightarrow +\infty$$

Then using $(\mathcal{B}')$, we deduce that $y = 0$.

3. Prescribed time stability

Let us consider the following non-autonomous bilinear system:

$$\begin{cases} \dot{z} = \mathcal{A}(t)z + \vartheta(t)B(t)z, & t \geq t_0 \\ z(t_0) = z_0 \in H \end{cases} \quad (3.1)$$

where $\{\mathcal{A}(t)\}_{t \geq t_0}$ is a family of linear bounded dissipative operators from V to V' , the family of linear bounded operators $\{B(t)\}_{t \geq t_0}$ of H is such that $\|B(\cdot)\| \in L^\infty(t_0, +\infty)$ and ϑ is a scalar control.

The objective is to design a scalar (time-varying) feedback control ϑ such that the respective closed-loop system has a good property in terms of stability.

Let us consider the following definitions of finite-time stabilization.

Definition 3.1.

- System (3.1) is said to be finite-time stabilizable if, there exists a time-varying feedback control $\vartheta(t, z)$ for which the system (3.1) is Lyapunov stabilizable and for any initial state, there exists a time $T_1 = T_1(t_0, z_0)$ for which $z(t) = 0$ for all $t \geq T_1$. In this case the function $T^*(t_0, z_0) =: \inf\{t > t_0 / z(t) = 0\}$ is called the settling time function. Moreover, if $\sup_{t_0 \geq 0, z_0 \in H} T^*(t_0, z_0) < +\infty$, then we have the so called fixed-time stabilization.
- We define the system (3.1) as prescribed-time stabilizable if, for any $T > 0$, there exists a time-varying feedback control $\vartheta(t, z)$ such that the system (3.1) is Lyapunov stabilizable and $\lim_{t \rightarrow (t_0+T)^-} z(t) = 0$.

In order to study the prescribed-time stability of system (3.1), we perform a variable change to reduce it to the previous case, i.e., the asymptotic case.

Let us consider, for a given $T > 0$, the following variable change:

$$y(t) = z(\gamma(t) + t_0), \quad \text{with } \gamma(t) = \frac{Tt}{t+1}. \quad (3.2)$$

For the sake of simplicity, and without loss of generality, we proceed with the assumption that $t_0 = 0$.

In this part, we suppose that the family of linear bounded operators $\{B(t)\}_{t \geq t_0}$ satisfies the following observability inequality:

$$\begin{aligned} \exists \tau, \delta > 0, \forall t_1 \geq t_0, \int_{t_1}^{t_1+\tau} |\langle B(\gamma(t) + t_0)\Phi_{t_1}(t), \Phi_{t_1}(t) \rangle| dt \\ \geq \delta \|\Phi_{t_1}(t_1 + \tau)\|^2, \quad \forall \Phi_{t_1} \in H \end{aligned} \quad (\mathcal{B}'')$$

where Φ_{t_1} is the solution of the homogeneous system

$$\begin{cases} \dot{\Phi}_{t_1} = \gamma'(t)\mathcal{A}(\gamma(t) + t_0)\Phi_{t_1}, & t \geq t_1 \\ \Phi_{t_1}(t_1) = \Phi_{t_1} \end{cases}$$

Theorem 3.1. *Let $T > 0$, and assume that $(\mathcal{B}'')$ holds, and that the operators $\{\mathcal{A}(t)\}_{t \geq t_0}$ satisfy the assumptions (H_1) , (H_2) . Then, for all $y_0 \in H$, the system (3.1) under the control defined for all $t \in [t_0, t_0 + T)$ by:*

$$\vartheta(t) = \vartheta(t, z) = \begin{cases} -\frac{T\langle B(t)z, z \rangle}{(T+t_0-t)^2 \|z\|^2}, & \text{if } z \neq 0 \\ 0, & \text{otherwise} \end{cases} \quad (3.3)$$

is finite time stable in the prescribed time T .

Proof. As mentioned earlier, we will prove Theorem 3.1 for $t_0 = 0$; the extension to the general case follows similarly.

The utilization of time-varying blowing-up functions enables us to demonstrate the well-posedness of the system (3.1) for times $t \in [0, \tilde{T}]$, where $\tilde{T}$ is any positive value less than T . Since boundedness holds until, but not including, the terminal time T , the well-posedness result we present is applicable only up to times arbitrarily close to T .

Now, taking into account that assumptions (H_1) , (H_2) are satisfied and that $F(t, z) = -\frac{T\langle B(t)z, z \rangle}{(T-t)^2 \|z\|^2} B(t)z$ satisfies (2.2) (with $K = \frac{5T}{(T-\tilde{T})^2} \|B\|$), we deduce according to Theorem 2.1 that system (3.1) admits a unique solution z on $[0, \tilde{T}]$ for any $0 < \tilde{T} < T$.

To study the prescribed time stability for the system (3.1), we consider for some $T > 0$, the following variable change

$$y(t) = z(\gamma(t)), \quad \text{with } \gamma(t) = \frac{Tt}{t+1},$$

which leads to the following system:

$$\begin{cases} \dot{y} = A(t)y + u(t, y)B(t)y, & t \geq 0 \\ y(0) = z_0 \in H \end{cases} \quad (3.4)$$

with $A(t) = \gamma'(t)\mathcal{A}(\gamma(t))$, $B(t) = B(\gamma(t))$ and $u(t, y)$ is given by (2.10).

The prescribed time stability of (3.1) is equivalent to the stability in infinite time of system (3.4). Taking into account the assumption $(\mathcal{B}'')$, we deduce from Theorem 2.2 and Remark 2.1 that system (3.4) is uniformly exponentially stable, so that

$$\lim_{t \rightarrow T^-} z(t) = \lim_{t \rightarrow +\infty} y(t) = 0.$$

Remark 3.1. Let us note that, from (2.22), we have the following estimate for some $M, \omega > 0$:

$$\|z(t)\|^2 \leq M \exp(-\omega(\frac{T}{T-t+t_0} - t_0)) \|z_0\|^2 \quad \forall t_0 \leq t \leq T + t_0. \quad (3.5)$$

In the next corollary, we introduce a necessary condition for the prescribed time stability of (3.1) under the control (3.3).

Proposition 1. *Let assumption (H_3) holds. If the system (3.1) is prescribed time stabilizable under the control (3.3) such that (3.5) holds, then there exist $\tau, \delta > 0$ such that the assumption $(\mathcal{B}'')$ holds.*

Proof. Performing the variable change given by (3.2), we deduce that y is the solution of the following system:

$$\begin{cases} \dot{y} = A(t)y + u(t, y)B(t)y, & t \geq 0 \\ y(0) = z_0 \in H \end{cases} \quad (3.6)$$

with $A(t) = \gamma'(t)\mathcal{A}(\gamma(t) + t_0)$, $B(t) = B(\gamma(t) + t_0)$ and $u(t, y)$ is given by (2.10). From (3.5) we deduce that

$$\begin{aligned} \|y(t)\|^2 &= \|z(\gamma(t) + t_0)\|^2 \\ &\leq M' \exp(-\sigma(t - t_0)) \|y_0\|^2, \quad \forall t \geq t_0, \end{aligned}$$

for some $M', \sigma > 0$, so we can deduce that system (3.6) is uniformly exponentially stabilizable. Taking into account that (H_3) holds we deduce from Theorem 2.3 that $(\mathcal{B}'')$ holds.

Remark 3.2.

- Taking into consideration the estimate (3.5), we can see that

$$\lim_{t \rightarrow (t_0+T)^-} \|\vartheta(t, z)B(t)z\| = 0.$$

Consequently, we can infer that the system can be extended beyond $T + t_0$ with a zero solution.

- In the light of the first point, we can deduce that prescribed-time stability implies fixed-time stability, which in turn implies finite-time stability.

4. Examples

4.1. Uniform exponential stability of a diffusion equation

The standard diffusion equation describes the diffusion of a quantity in terms of its spatial distribution. The non-autonomous diffusion equation extends this concept by allowing the coefficients of the equation, such as diffusivity $\alpha(t)$, to vary with time. Our goal here, is to study the stability of the following non-autonomous diffusion bilinear system:

$$\begin{cases} y_t = \alpha(t)\Delta y + \vartheta(t, y)\beta(t)\chi_\omega y, & \text{in } Q = (0, 1) \times (0, +\infty) \\ y(0, t) = y(1, t) = 0, & \text{in } (0, +\infty) \\ y(x, 0) = y_0 \in H, & \text{in } (0, 1) \end{cases} \quad (4.1)$$

where $\omega = (a, b)$, with $0 \leq a < b \leq 1$, and χ_ω is the restriction operator on the set ω . The functions $\alpha, \beta \in L^2(0, +\infty)$ satisfy, for a given constant $c > 0$: $\alpha(t) \geq c$ and $|\beta(t)| \geq c\alpha(t)$ for all $t \geq 0$. The function $y(t)$ represents the state and $\vartheta(t, y)$ is the (time-varying) feedback control given by (2.10).

The system (4.1) takes the form of (2.1) by taking $\mathcal{A}(t) = \alpha(t)A$ with $A = \Delta$ and $D(\mathcal{A}(t)) = D(A) = H_0^1(0, 1) \cap H^2(0, 1)$, $B(t) = \beta(t)\chi_\omega$ and $t_0 = 0$. Moreover given $T > 0$, we can see that assumptions (H_1) , (H_2) are satisfied for $H = L^2(0, 1)$ and $V = H_0^1(0, 1)$.

Now, we need to check that the observability condition (B) is verified.

For $t_1 > 0$, let us consider the homogeneous system

$$\begin{cases} \dot{\varphi}_t = \alpha(t)A\varphi_t, & t \geq t_1 \\ \varphi_{t_1}(t_1) = \varphi_1 \end{cases} \quad (4.2)$$

The solution of (4.2) is given by

$$\varphi_{t_1}(t) = S(F(t) - F(t_1))\varphi_1,$$

where F is the primitive function of α and $S(t)$ is the semigroup generated by the operator A .

Observing that $|\beta(t)| \geq c\alpha(t)$, we deduce that for $\tau > 0$, we have

$$\begin{aligned} & \int_{t_1}^{t_1+\tau} |\langle B(t)\varphi_{t_1}(t), \varphi_{t_1}(t) \rangle| dt \\ &= \int_{t_1}^{t_1+\tau} \int_a^b |\beta(t)| \left([\varphi_{t_1}(t)](x) \right)^2 dx dt \\ &= \int_{t_1}^{t_1+\tau} |\beta(t)| \int_a^b \left((S(F(t) - F(t_1))\varphi_1)(x) \right)^2 dx dt \\ &\geq \int_{t_1}^{t_1+\tau} c\alpha(t) \int_a^b \left((S(F(t) - F(t_1))\varphi_1)(x) \right)^2 dx dt \end{aligned}$$

Performing the following variable change $s = F(t) - F(t_1)$, we deduce that

$$\begin{aligned} & \int_{t_1}^{t_1+\tau} |\langle B(t)\varphi_{t_1}(t), \varphi_{t_1}(t) \rangle| dt \\ &\geq c \int_0^{c\tau} \int_a^b ((S(s)\varphi_1)(x))^2 dx ds \end{aligned}$$

According to (Rosier, 2007, p. 391), we deduce that

$$\begin{aligned} \int_{t_1}^{t_1+\tau} |\langle B(t)\varphi_{t_1}(t), \varphi_{t_1}(t) \rangle| dt &\geq \delta \|S(F(t_1 + \tau) - F(t_1))\varphi_1\|^2 \\ &= \delta \|\varphi_{t_1}(t_1 + \tau)\|^2 \end{aligned}$$

for some $\delta > 0$.

Taking into account that the assumptions (H_1) , (H_2) also hold, we deduce according to Theorem 2.2 that the system (4.1) is uniformly exponentially stable under the following control

$$u(t, y) = \begin{cases} -|\beta(t)| \frac{\|\chi_\omega y\|^2}{\|y\|^2}, & \text{if } y \neq 0 \\ 0, & \text{otherwise} \end{cases} \quad (4.3)$$

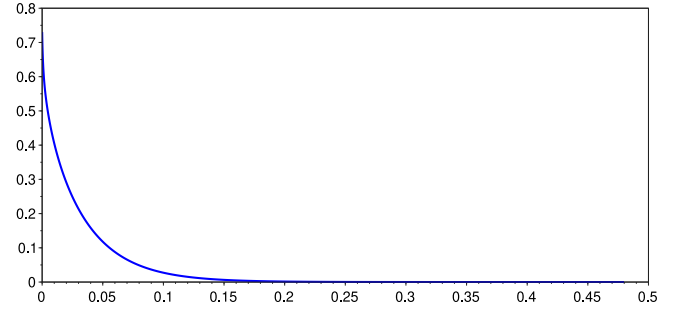


Fig. 1. The evolution of the norm of the state $\|y\|$.

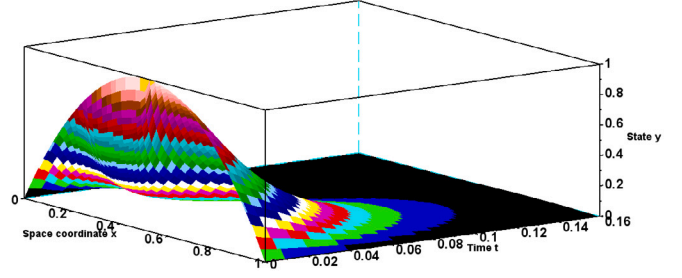


Fig. 2. The evolution of the state $y(x, t)$ w.r.t to x and time t .

For the numerical approach, we applied the finite difference method for spatial discretization, a popular numerical technique that simplifies partial differential equations (PDE's) into a system of ordinary differential equations (ODE's). This method is particularly useful for its ease of implementation and computational efficiency, especially when dealing with problems in bounded domains such as $\Omega = (0, 1)$.

Key Parameters for Numerical Simulations:

- Domain $\Omega = (0, 1)$, control region $\omega = [0.25, 0.5]$, $\alpha(t) = 0.1 + \frac{1}{t+1}$ and $\beta(t) = 30 * \alpha(t)^2$ for $t \geq 0$.
- Discretization intervals: time step: $\delta t = 1.6 \times 10^{-4}$, and space step: $\delta x = 0.02$.
- Initial conditions: $y_0(x) = 4x(1 - x)$.

Results Summary:

- Fig. 1: Displays the evolution of the norm of the state $\|y\|$, which converges to 0, illustrating the stability.
- Fig. 2, shows the 3d-evolution of the state w.r.t space coordinate x and time t .

4.2. Finite time stability of transport equation

In this section, our goal is to study the prescribed-time stability of a bilinear transport equation with mobile control support. This scenario often arises in practical applications, where there is a need to influence the transport process within certain spatial regions and dynamically adjust this influence over time. Let us consider the following system:

$$\begin{cases} y_t = -y' + \vartheta(t, y)\chi_{\omega(t)}y & \text{in } Q = (0, +\infty)^2 \\ y(0, t) = 0 & \text{in } (0, +\infty) \\ y(x, 0) = y_0 \in H & \text{in } (0, +\infty), \end{cases} \quad (4.4)$$

where,

- $y(x, t)$ represents the quantity being transported, and x being a spatial coordinate and t is time;

- y_t denotes the partial derivative of y with respect to time t , while y' represents the partial derivative of y with respect to the spatial coordinate x ;
- $\vartheta(t, y)$ is the (time-varying) feedback control

$$\vartheta(t, y) = \begin{cases} -\frac{T\|\chi_{\omega(t)}y\|^2}{(T-t)^2\|y\|^2}, & \text{if } y \neq 0 \\ 0, & \text{otherwise} \end{cases}$$

- $\chi_{\omega(t)}$ is the characteristic function of the set $\omega(t)$, which represents the moving support;
- $\omega(t)$ is given by $\omega(t) = (f(t), +\infty)$, such that for some $\tau > 0$, $f(t) = (\tau - t)^2$ for all $t > 0$;
- $Q = (0, +\infty)^2$ represents the spatial-temporal domain over which the equation is defined.

Taking $\mathcal{A}(t) = -\frac{\partial}{\partial x}$ with $D(\mathcal{A}(t)) = \{y \in H^1(0, +\infty) / y(0) = 0\} = H_0^1(0, +\infty)$, (which generates a semigroup of translation $S(t)$) and $\mathcal{B}(t) = \chi_{\omega(t)}$, the system (4.4) takes the form of (3.1). Taking $H = L^2(0, +\infty)$ and $V = H_0^1(0, +\infty)$. The wellposedness of the system is ensured by Pazy (2012). Let us now show that the observability condition (B'') is verified.

Let us consider the homogeneous system

$$\begin{cases} \dot{\Phi}_1 = -\gamma'(t) \frac{\partial \Phi_1}{\partial x}, & t \geq t_1 \\ \Phi_1(t_1) = \Phi_1 \end{cases} \quad (4.5)$$

with $\Phi_1 \in H$ and $\gamma(t) = \frac{\tau}{t+1}$.

The solution of (4.5) is given by

$$\Phi_{t_1}(t) = S(\gamma(t) - \gamma(t_1))\Phi_1,$$

where the semigroup of translation $S(t)$ is given by

$$[S(t)\Phi_1](x) = \begin{cases} \Phi_1(x-t), & \text{if } x-t \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (4.6)$$

So that

$$\Phi_{t_1}(x, t) = \begin{cases} \Phi_1(x - \gamma(t) + \gamma(t_1)), & \text{if } x \geq \gamma(t) - \gamma(t_1) \\ 0, & \text{otherwise} \end{cases} \quad (4.7)$$

$$\begin{aligned} & \int_{t_1}^{t_1+2\tau} |\langle \mathcal{B}(\gamma(t))\Phi_{t_1}(t), \Phi_{t_1}(t) \rangle| dt \\ &= \int_{t_1}^{t_1+2\tau} \|\chi_{\omega(t)}\Phi_{t_1}(t)\|^2 dt \\ &= \int_{t_1}^{t_1+2\tau} \int_{f(\gamma(t))}^{+\infty} \Phi_{t_1}^2(t) dx dt \\ &= \int_{t_1}^{t_1+2\tau} \int_{\sup(f(\gamma(t)), \gamma(t)-\gamma(t_1))}^{+\infty} \Phi_1^2(x - \gamma(t) + \gamma(t_1)) dx dt \\ &\geq \int_{t_1+\tau}^{t_1+2\tau} \int_{\sup(f(\gamma(t)), \gamma(t)-\gamma(t_1))}^{+\infty} \Phi_1^2(x - \gamma(t) + \gamma(t_1)) dx dt \end{aligned}$$

Moreover, one can easily verify that $f(\gamma(t)) \leq \gamma(t) - \gamma(t_1)$ for all $t_1 + \tau \leq t \leq t_1 + 2\tau$, thus

$$\begin{aligned} & \int_{t_1}^{t_1+2\tau} |\langle \mathcal{B}(\gamma(t))\Phi_{t_1}(t), \Phi_{t_1}(t) \rangle| dt \\ &\geq \int_{t_1+\tau}^{t_1+2\tau} \int_{\gamma(t)-\gamma(t_1)}^{+\infty} \Phi_1^2(x - \gamma(t) + \gamma(t_1)) dx dt \end{aligned}$$

Then, we obtain via a variable change

$$\begin{aligned} & \int_{t_1}^{t_1+2\tau} |\langle \mathcal{B}(\gamma(t))\Phi_{t_1}(t), \Phi_{t_1}(t) \rangle| dt \\ &\geq \int_{t_1+\tau}^{t_1+2\tau} \int_0^{+\infty} \Phi_1^2(x) dx dt \\ &\geq \tau \|\Phi_1\|^2 = \tau \|\Phi_{t_1}(t_1 + 2\tau)\|^2. \end{aligned}$$

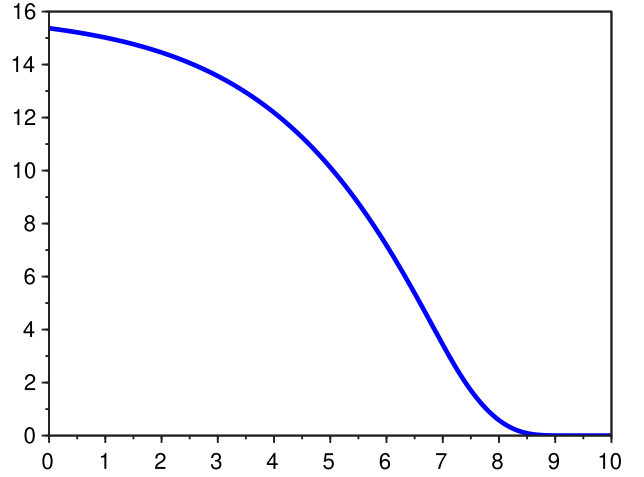


Fig. 3. The evolution of the norm of the state $\|y\|$.

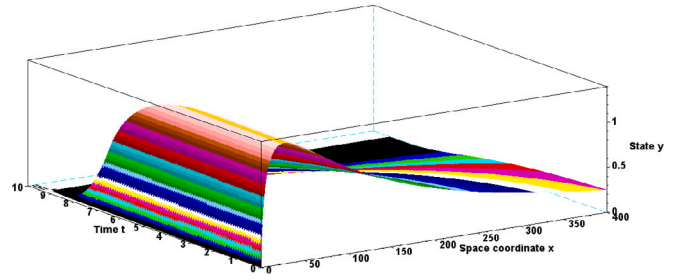


Fig. 4. The evolution of the state $y(x, t)$ w.r.t to x and time t .

We deduce from Theorem 3.1 that the system (4.4) is prescribed stable at the time τ . Now, taking into account Remark 3.2, we deduce that the system is stable at any $T \geq \tau$, and hence the system (4.4) is fixed time stable.

Key Parameters for Numerical Simulations:

- Domain $\Omega = (0, 2000)$, $T = \tau = 10$.
- Discretization intervals: time step: $\delta t = 0.05$, and space step: $\delta x = 0.1$.
- Initial conditions: $y_0(x) = 4x(1-x)$.

Results Summary:

- Fig. 3: Displays the evolution of the norm of the state $\|y\|$, which converges to 0, illustrating the stability.
- Fig. 4, shows the 3d-evolution of the state w.r.t space coordinate x and time t .

5. Conclusion

We have introduced (time-varying) feedbacks laws to achieve the uniform exponential and prescribed time stabilization of a class of non-autonomous bilinear systems. The uniform exponential stability is proved through the computation of the time derivative of the system's "energy", and using a specific weak observability inequality. As for the stability in finite-time, we applied the previous results on asymptotic stability to derive the prescribed-time stability using a time-varying feedback control. These results are subsequently applied to the diffusion and transport equations. Moving forward, there are several promising directions for future research. One focus could be on making the system more robust to uncertainties and improving energy

efficiency in real applications. Another challenge is extending the stabilization techniques to nonlinear systems, which requires adjustments to handle their complex behavior. Additionally, studying the stabilization of non-autonomous systems with unbounded control operators is an interesting direction that we plan to explore in future work.

CRedit authorship contribution statement

Hanan Najib: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Conceptualization. **Soufiane Yahyaoui:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Mohamed Ouzahra:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Ammari, K., El Alaoui, S., & Ouzahra, M. (2021). Feedback stabilization of linear and bilinear unbounded systems in Banach space. *Systems & Control Letters*, 155, Article 104987.
- Anderson, B. D., Ilchmann, A., & Wirth, F. R. (2013). Stabilizability of linear time-varying systems. *Systems & Control Letters*, 62(9), 747–755.
- Astolfi, A., Karagiannis, D., & Ortega, R. (2008). *Nonlinear and adaptive control with applications: Vol. 187*, (p. 14). London: Springer.
- Ball, J. M., & Slemrod, M. (1979). Feedback stabilization of distributed semilinear control systems. *Applied Mathematics and Optimization*, 5, 169–179.
- Barreira, L. (2008). *Stability of nonautonomous differential equations*. Springer-Verlag.
- Barreira, L., & Valls, C. (2005). Stability of nonautonomous differential equations in Hilbert spaces. *Journal of Differential Equations*, 217(1), 204–248.
- Berrahmoune, L. (2010). Stabilization of unbounded bilinear control systems in Hilbert space. *Journal of Mathematical Analysis and Applications*, 372(2), 645–655.
- Bhat, S. P., & Bernstein, D. S. (2000). Finite-time stability of continuous autonomous systems. *SIAM Journal on Control and Optimization*, 38(3), 751–766.
- Chen, P. C., & Huang, A. C. (2005). Adaptive sliding control of non-autonomous active suspension systems with time-varying loadings. *Journal of Sound and Vibration*, 282(3–5), 1119–1135.
- Corduneanu, C. (1977). *Principles of differential and integral equations: Vol. 295*, American Mathematical Soc.
- Dautray, R., & Lions, J. L. (1992). *Mathematical analysis and numerical methods for science and technology: evolution problems I: Vol. 5*, Springer Science & Business Media.
- El Alaoui, S., & Ouzahra, M. (2024). Exponential stability of linear systems under a class of Desch–Schappacher perturbations. *Mathematical Methods in the Applied Sciences*, 47(4), 2753–2768.
- El Ayadi, R. E., Ouzahra, M., & Boutoulout, A. (2012). Strong stabilisation and decay estimate for unbounded bilinear systems. *International Journal of Control*, 85(10), 1497–1505.
- Espitia, N., Polyakov, A., Efimov, D., & Perruquetti, W. (2019). Boundary time-varying feedbacks for fixed-time stabilization of constant-parameter reaction–diffusion systems. *Automatica*, 103, 398–407.
- Jammazi, C., Boutayeb, M., & Saidi, K. (2023). On the fixed-time extinction based nonlinear control and systems decomposition: applications to bilinear systems. *Chaos, Solitons & Fractals*, 174, Article 113893.
- Jammazi, C., Ouzahra, M., & Sogoré, M. (2023). Small-time extinction with decay estimate of bilinear systems on Hilbert space. *Journal of Nonlinear Science*, 33(4), 54.
- Kalman, R. E. (1960). Contributions to the theory of optimal control. *Boletín de la Sociedad Matemática Mexicana*, 5(2), 102–119.
- Li, M., Jiang, M., Ren, F., & Zhang, Y. (2022). Finite-time stability of a class of nonautonomous systems. *International Journal of Control*, 95(10), 2771–2779.
- Li, W., Xie, Z., Wong, P. K., Ma, X., Cao, Y., & Zhao, J. (2019). Nonfragile H_∞ control of delayed active suspension systems in finite frequency under nonstationary running. *Journal of Dynamic Systems, Measurement, and Control*, 141(6), Article 061001.
- Li, X., Yang, X., & Song, S. (2019). Lyapunov conditions for finite-time stability of time-varying time-delay systems. *Automatica*, 103, 135–140.
- Lopez-Ramirez, F., Efimov, D., Polyakov, A., & Perruquetti, W. (2019). Conditions for fixed-time stability and stabilization of continuous autonomous systems. *Systems & Control Letters*, 129, 26–35.
- Moulay, E., & Perruquetti, W. (2008). Finite-time stability conditions for non-autonomous continuous systems. *International Journal of Control*, 81(5), 797–803.
- Najib, H., & Ouzahra, M. (2024a). Output finite-time stabilisation of a class of linear and bilinear control systems. *International Journal of Control*, 97(4), 690–700.
- Najib, H., & Ouzahra, M. (2024b). Stabilization in finite time of a class of unbounded non-linear systems. *Journal of Dynamical and Control Systems*, 1–17.
- Ouzahra, M. (2020). Uniform exponential stabilization of nonlinear systems in Banach spaces. *Mathematical Methods in the Applied Sciences*, 43(12), 7311–7325.
- Ouzahra, M. (2021). Finite-time control for the bilinear heat equation. *European Journal of Control*, 57, 284–293.
- Pazy, A. (2012). *Semigroups of linear operators and applications to partial differential equations: Vol. 44*, Springer Science & Business Media.
- Polyakov, A. (2012). Nonlinear feedback design for fixed-time stabilization of linear control systems. *IEEE Transactions on Automatic Control*, 57(8), 2106–2110.
- Polyakov, A., Coron, J. M., & Rosier, L. (2017). On boundary finite-time feedback control for heat equation. *IFAC-PapersOnLine*, 50(1), 671–676.
- Polyakov, A., Coron, J. M., & Rosier, L. (2018). On homogeneous finite-time control for linear evolution equation in Hilbert space. *IEEE Transactions on Automatic Control*, 63(9), 3143–3150.
- Polyakov, A., Efimov, D., & Perruquetti, W. (2015). Finite-time and fixed-time stabilization: Implicit Lyapunov function approach. *Automatica*, 51, 332–340.
- Rosier, L. (2007). A survey of controllability and stabilization results for partial differential equations. *Journal Européen Des Systemes Automatisés*, 41(3/4), 365.
- Silverman, L. M., & Meadows, H. E. (1967). Controllability and observability in time-variable linear systems. *SIAM Journal on Control*, 5(1), 64–73.
- Sogoré, M., & Jammazi, C. (2020). On the global finite-time stabilization of bilinear systems by homogeneous feedback laws. Applications to some PDE's. *Journal of Mathematical Analysis and Applications*, 486(2), Article 123815.
- Song, Y., Wang, Y., Holloway, J., & Krstic, M. (2017). Time-varying feedback for regulation of normal-form nonlinear systems in prescribed finite-time. *Automatica*, 83, 243–251.
- Song, Y., Wang, Y., & Krstic, M. (2019). Time-varying feedback for stabilization in prescribed finite-time. *International Journal of Robust and Nonlinear Control*, 29(3), 618–633.
- Xu, Y., Xie, Z., Zhao, J., Li, W., Li, P., & Wong, P. K. (2021). Robust non-fragile finite frequency H_∞ control for uncertain active suspension systems with time-delay using TS fuzzy approach. *Journal of the Franklin Institute*, 358(8), 4209–4238.
- Zhang, W., Han, Q. L., Tang, Y., & Liu, Y. (2019). Sampled-data control for a class of linear time-varying systems. *Automatica*, 103, 126–134.
- Zhang, K. K., Zhou, B., & Duan, G. R. (2024). Global prescribed-time output feedback control of a class of uncertain nonlinear systems by linear time-varying feedback. *Automatica*, 165, Article 111680.
- Zhou, B. (2016). On asymptotic stability of linear time-varying systems. *Automatica*, 68, 266–276.
- Zhou, B. (2021). Lyapunov differential equations and inequalities for stability and stabilization of linear time-varying systems. *Automatica*, 131, Article 109785.
- Zhou, B., & Shi, Y. (2021). Prescribed-time stabilization of a class of nonlinear systems by linear time-varying feedback. *IEEE Transactions on Automatic Control*, 66(12), 6123–6130.